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AI-Driven Approach for Enhancing Sustainability in Urban Public Transportation

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Abstract: The functioning of modern urban environments relies heavily on the public transport system. Given spatial, economic, and sustainability criteria, public transport in larger urban areas is unrivaled. The system's role depends on the quality of service it offers. Achieving the desired service quality requires a design that meets transport demands. This paper uses a data-driven approach to address headway deviations in public transport lines and explores ways to improve regularity during the design phase. Headway is a critical dynamic element for transport organization and passenger quality. Deviations between planned and actual headways represent disturbances. On lines with headways under 15 min, passengers typically do not consult schedules, making punctuality less crucial. Reduced headway regularity affects the average travel time, travel time uncertainty, and passenger comfort. Ideally, the public transport system operates with regular headways. However, disturbances can spread and affect subsequent departures, leading to vehicle bunching. While previous research focused on single primary disturbances, this study, with the help of AI (reinforcement learning), examines multiple primary disturbances in the cities of Belgrade, Novi Sad, and Niš. The goal is to model the cumulative impact of these disturbances on vehicle movement. By ranking parameter influences and using the automatic optimization of static line elements, this research aims to improve headway regularity and increase system resilience to disturbances. The results of this research could also be useful in developing adaptive public transport management systems that leverage AI and IoT technologies to continuously optimize headway regularity in response to real-time data, ultimately enhancing service quality and passenger satisfaction.

Keywords: AI; sustainability; public transport; sustainable transportation

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1. Introduction

Urban public transportation systems are integral to the functionality and sustainability of modern cities, serving as a critical link in the urban mobility ecosystem. The advent of Artificial Intelligence (AI) has introduced transformative capabilities, enabling significant enhancements in operational efficiency and service quality [1]. This paper explores the

application of AI-driven approaches in optimizing public transportation systems, focusing on their role in improving scheduling, route planning, and predictive maintenance. By employing machine learning, real-time data analytics, and advanced optimization algorithms, AI addresses key challenges such as headway deviations, traffic congestion, and passenger flow management.

Through a literature review of case studies from various European cities [2–4], this research highlights how AI applications contribute to increased system resilience, reduced operational costs, and enhanced passenger satisfaction. Understanding these advancements offers valuable insights into the potential of AI to drive sustainable improvements in urban public transport [5]. It must be stated that efficient public transport depends on many conditions and the functioning of cooperating systems such as park-and-ride and bike-sharing systems [6].

Existing gaps in the literature suggest the need to examine the critical factors that influence the success of AI implementations, such as data quality, algorithm selection, and integration strategies. By doing so, it aims to provide a comprehensive framework for future research and practical applications in urban transportation planning.

The main research question is defined as follows:

RQ: "How can AI-driven models optimize headway regularity to improve the sustainability and efficiency of urban public transportation systems?"

By sustainability in urban public transportation, the authors of this paper refer to creating a system that efficiently meets current and future transportation needs while minimizing environmental impact, reducing resource consumption, and promoting social equity through the integration of advanced technologies like AI.

This research will investigate how AI-driven models can optimize headway regularity to enhance the sustainability and efficiency of urban public transportation. Key factors include the effectiveness of AI algorithms, data integration, optimization strategies, sustainability metrics, operational efficiency, passenger experience, implementation challenges, and insights from case studies.

In the subsequent sections, this paper delves into the existing literature on this topic, presenting key theoretical findings relevant to sustainable, data-driven enhancements in public transport services. The methodological framework for applying AI in these processes has been examined through a sample of urban transport companies in three cities from Serbia, and it is presented in Section 3, accompanied by a breakdown of the research instruments used. Section 4 then presents the results of the quantitative research, including descriptive statistics, and the identification of key factors and enhancements made using AI.

Section 5 discusses the main findings, compares them with similar research, and engages in a broader discussion with relevant previous works in the literature. This paper concludes with Section 6, by outlining future research plans, emphasizing the ongoing importance of understanding and optimizing data-driven approaches to improvements in urban transport.

2. Materials and Methods

Artificial Intelligence (AI) is revolutionizing urban public transportation by enhancing operational efficiency and service quality. This section explores the application of AI technologies in optimizing various aspects of public transport systems, including scheduling, route planning, and predictive maintenance. By leveraging machine learning algorithms, real-time data analytics, and predictive modeling, AI addresses critical challenges such as headway deviations, traffic management, and passenger flow [6,7].

Case studies from diverse global contexts highlight the transformative impact of AI, illustrating improvements in service regularity, operational reliability, and passenger satisfaction. Understanding these applications provides a foundation for assessing AI's role in advancing sustainable urban transport solutions [8].

2.1. Existing AI Applications in Public Transportation

The integration of Artificial Intelligence (AI) into public transportation systems has revolutionized how cities manage and optimize transit operations. Current AI technologies and methodologies encompass a range of applications aimed at improving scheduling, route planning, predictive maintenance, traffic management, and passenger flow optimization [9,10].

One of the most significant AI applications in public transportation is the optimization of scheduling and route planning. Machine learning algorithms, particularly those employing neural networks and reinforcement learning, are being used to analyze vast amounts of data, including historical ridership, traffic patterns, and real-time transit information. These algorithms can dynamically adjust schedules and routes to enhance efficiency and reduce delays, ensuring that public transport services are more reliable and punctual [11].

Predictive maintenance is another critical area where AI has made substantial inroads. By utilizing AI-powered predictive analytics and anomaly detection techniques, transportation agencies can monitor the sound and efficiency of their vehicles and infrastructure in real time [12,13]. On the other side, this allows for the early identification of potential issues, enabling proactive maintenance which minimizes downtime and extends the lifespan of transit assets. As a result, operational costs are reduced, and the overall safety and reliability of public transportation systems are improved [14].

AI is also instrumental in managing traffic signals and prioritizing public transport vehicles. Adaptive signal control systems use AI to analyze real-time traffic conditions and adjust signal timings to minimize delays for buses and trams. This not only improves the punctuality of public transport but also helps in reducing traffic congestion and emissions, contributing to more sustainable urban environments [15,16].

Addressing headway deviations is a critical aspect of improving the regularity and reliability of public transport lines. Research indicates that headway deviations—variations in the time intervals between successive vehicles—significantly impact service quality and passenger satisfaction. In the design phase, several strategies can be implemented to mitigate these deviations [17,18].

One effective approach is the use of AI-driven optimization techniques. Machine learning algorithms can analyze historical data on vehicle arrivals and departures, identifying patterns and predicting potential deviations. By understanding these patterns, transit agencies can design schedules that are more resilient to disruptions [19]. For example, reinforcement learning can be applied to dynamically adjust vehicle dispatch times, ensuring more consistent headways even under varying traffic conditions [20].

Simulation models are also employed to test different scheduling scenarios during the design phase. These models incorporate factors such as traffic flow, passenger demand, and vehicle performance to evaluate the impact of different scheduling strategies on headway regularity. By simulating various conditions, planners can identify optimal schedules that minimize deviations and enhance service reliability [21–23].

Moreover, integrating real-time data analytics into the design phase can significantly improve headway regularity. Real-time data from GPS, automated vehicle location systems, and passenger counting systems provide insights into actual vehicle performance and passenger behavior. These data can be used to adjust schedules proactively, reducing the gap between planned and actual headways [24–26].

In addition to these technical solutions, infrastructural adjustments during the design phase can also help in maintaining regular headways. Dedicated bus lanes, priority signaling at intersections, and the strategic placement of bus stops are some measures that can minimize delays and ensure a smoother vehicle flow [27,28].

Collaborative efforts between transportation planners, data scientists, and urban developers are essential in addressing headway deviations. By leveraging AI technologies and 5G data, real-time data, and robust simulation models, public transport systems can be designed to achieve more regular headways, thereby improving the overall service quality and passenger satisfaction [29–31].

Passenger flow management has seen significant advancements through the application of AI. Computer vision and data analytics are employed to monitor and predict passenger movements within transit hubs and vehicles. This information is used to optimize the deployment of resources, such as adjusting the frequency of services during peak times or managing crowd-control measures during special events. Enhanced passenger flow management leads to improved travel experiences, reduced wait times, and increased safety for commuters [32,33].

Real-time information systems powered by AI provide passengers with up-to-date information on transit schedules, delays, and alternative routes. These systems enhance the overall user experience by allowing passengers to make informed decisions about their travel plans. Additionally, AI-driven chatbots and virtual assistants are being deployed to assist passengers with queries and provide support, further enhancing the accessibility and convenience of public transportation [34,35].

Overall, the application of AI technologies in public transportation systems is driving significant improvements in efficiency, reliability, and sustainability. By leveraging advanced methodologies and real-time data analytics, AI is transforming how cities manage their transit operations, ultimately leading to a better service quality and enhanced urban mobility [36–38].

Case studies and research on AI-driven optimization in public transportation reveal significant advancements in scheduling, route planning, predictive maintenance, and passenger flow management [39,40]. In scheduling and route planning, several cities have successfully implemented AI algorithms to enhance service reliability and efficiency. For instance, cities like Vienna and Lisbon have utilized machine learning models to optimize bus and tram schedules, resulting in reduced wait times and improved adherence to timetables [41–44]. These models analyze vast datasets, including historical ridership and real-time traffic conditions, to dynamically adjust routes and schedules.

In predictive maintenance, AI applications have demonstrated their potential to significantly reduce operational costs and improve safety. Cities such as Madrid and London have adopted AI-powered predictive maintenance systems that use sensors and data analytics to monitor the health of transit vehicles and infrastructure [45–47]. These systems predict potential failures before they occur, allowing for timely maintenance and minimizing downtime.

Passenger flow management has also benefited from AI-driven solutions. For example, in Barcelona, AI models have been employed to analyze passenger movement patterns using data from ticketing systems and sensors. This information is used to optimize service frequency and manage crowding, particularly during peak hours. The result is a more efficient and comfortable travel experience for passengers [48–50].

AI has significantly enhanced service quality, operational efficiency, and passenger satisfaction in public transportation. By optimizing scheduling and route planning, AI ensures more reliable and punctual services, reducing wait times for passengers. Predictive maintenance powered by AI minimizes vehicle downtime and operational disruptions, enhancing reliability and safety [51–53].

AI-driven traffic signal control and passenger flow management further streamline operations, leading to smoother travel experiences. Case studies from cities like Vienna and Barcelona illustrate these benefits, showing increased passenger satisfaction due to reduced congestion and improved service regularity. Overall, AI's impact results in a more efficient, reliable, and user-friendly public transport system [54–56].

Now follows the display of all research factors that were analyzed during the empirical part of this research, using data from three major cities in Serbia.

2.2. Research Factors to Be Analyzed

To address the research question, several key research factors should be examined within the empirical part of this research:

1. AI Algorithms and Techniques:
   - Types of AI Models:
     - Investigate different AI models (e.g., machine learning, deep learning, reinforcement learning) used to optimize scheduling and headway regularity (also examined in [57]).
   - Algorithm Efficiency:
     - Assess the computational efficiency and scalability of various AI algorithms in real-world urban transport scenarios.
2. Data Utilization and Integration:
   - Data Sources:
     - Identify the types of data (e.g., historical ridership, real-time traffic, weather conditions) required for effective AI modeling.
   - Data Quality and Integration:
     - Evaluate how data quality and integration from diverse sources impact the accuracy and reliability of AI models.
3. Optimization Strategies:
   - Headway Regularity Models [58]:
     - Analyze specific AI-driven optimization strategies for maintaining regular headways and minimizing deviations.
   - Impact of Dynamic Adjustments [59]:
     - Explore how real-time adjustments and predictive analytics contribute to optimizing headway regularity.
4. Sustainability Metrics (analyzed in use cases from [60–62]):
   - Resource Efficiency:
     - Assess the impact on operational resource use, such as vehicle and personnel efficiency, because of AI optimization.
5. Operational Efficiency [63,64]:
   - Service Reliability:
     - Measure improvements in service reliability and punctuality due to optimized headway regularity.
   - Cost–Benefit Analysis:
     - Conduct cost–benefit analyses to determine the economic advantages of implementing AI-driven optimization.
6. Passenger Experience [65]:
   - Travel Time and Comfort:
     - Investigate how enhanced headway regularity affects passenger travel time, comfort, and overall satisfaction.
   - User Acceptance:
     - Study the acceptance and perception of AI-driven scheduling changes among passengers.

In the next subchapter, the authors formulated the research hypothesis.

2.3. Research Hypothesis Formulation

From the defined research question and through a literature review, the authors identified two main research hypotheses that should be analyzed further via quantitative research.

Hypothesis 1. *Implementing AI-driven optimization models in urban public transportation systems will significantly reduce headway deviations, leading to improved service regularity and operational efficiency.*

Hypothesis 2. *AI-driven predictive maintenance and real-time data analytics will enhance the sustainability of urban public transportation by reducing vehicle downtime and operational costs, thereby increasing overall system resilience and passenger satisfaction.*

These hypotheses can guide research examining the effectiveness of AI technologies in achieving specific improvements in urban public transport systems.

Now follows the chapter about the methodological framework for this research.

3. Methodological Framework

3.1. Survey Description and Sample Definition

The sample included public urban transportation data from three major cities (Belgrade, Novi Sad, and Nis) in the Republic of Serbia (West Balkans region of Europe).

The authors focused on public buses as the most common way of transport in Serbia, also because only Belgrade has multiple means of public transport, such as trams, trolleys, trains, etc. Data were collected from the official databases of public enterprises for passenger transport services in the cities of Belgrade, Novi Sad, and Nis. Descriptive statistics for all the collected data are presented in Table 1.

Table 1. Profile of sampled public transport enterprises.

	Belgrade	Novi Sad	Nis
Number of employees	5850	1300	110
Number of vehicles for passenger transport (buses)	650	270	150
Average age of buses in operation	8	12	11
Annual revenue range (in mil EUR)	156	27	7

The authors approached existing, historic data about all 1.070 vehicles from the sampled three cities. Email addresses were derived from the available databases. The authors asked representatives from these enterprises whether the chief data or chief IT officer could deliver all necessary data for analysis. Since these public-owned enterprises are subject to external review, people in charge of the above departments for the three sampled enterprises delivered all available data in a timely manner.

3.2. Framework for Analysis

To effectively investigate the hypotheses related to AI-driven optimization and predictive maintenance in urban public transportation, it is necessary to establish a comprehensive research framework. This framework will ensure the validity and reliability of the conclusions drawn from this study. Key dimensions are defined in Table 2.

Table 2. Key dimensions for quantitative research.

Urban Public Transport Sustainability Dimension	Previously Analyzed in
Headway deviations	[66,67]
Service regularity	[68,69]
Operational efficiency	[70]
Vehicle downtime	[71,72]
Operational costs	[73,74]
System resilience	[75,76]
Passenger satisfaction	[77]

Available raw data from last five years—2019, 2020, 2021, 2022, and 2023—on urban public transportation systems in the three largest Serbian cities were collected, analyzed, and consolidated into a unique dataset. The data sources included real-time traffic information, vehicle performance metrics, and passenger flow data, provided by the city transportation departments and public transport enterprises via Excel files and, in case of the city of Belgrade, via the official transport management software v 2.0.

Initially, the authors conducted interviews with transportation planners and decision makers to determine the preliminary use of AI-driven approaches in their systems. These discussions helped identify whether cities employed predictive analytics to forecast potential disruptions based on historical data or focused on analyzing past performance to track key indicators of operational efficiency and service quality.

Following data collection and the preliminary interviews, a data standardization process was initiated. The dataset was thoroughly checked for quality, and two rounds of data alignment were conducted due to the differing sources and formats of the data. The authors then checked for initial correlations within the data to avoid multicollinearity and confirm that both AI-driven approaches (optimization models and predictive maintenance) were sufficiently correlated with the analyzed dimensions of service regularity, operational efficiency, sustainability, and passenger satisfaction. By ensuring data integrity and employing robust statistical techniques, this framework provided a solid foundation for evaluating the impact of AI on urban public transportation systems, leading to valid and actionable insights for improving sustainability and efficiency.

The authors employed reinforcement learning (RL) as an AI technique [61], which can learn from interactions with the environment, to make decisions that improve scheduling efficiency and reduce headway deviations. By continuously adjusting based on real-time feedback, RL can dynamically optimize routes and schedules, making it ideal for addressing the first hypothesis about reducing headway deviations and improving service regularity. The authors' focus was to build prototype algorithms to test research hypotheses. The goal was to implement RL algorithms and develop adaptive scheduling systems that learnt from historical data and real-time conditions to minimize headway deviations and enhance operational efficiency.

The first step was problem formulation, where, in the context of public transportation, the state space included various elements such as the current positions of vehicles, traffic conditions, passenger demand, and schedules. The action space consisted of possible decisions the system could make, such as adjusting the departure times, rerouting vehicles, or modifying service frequencies. The technique included the reward function, which measured the effectiveness of actions taken by the RL agent. In scheduling optimization, rewards could be based on criteria such as minimizing headway deviations, reducing wait times for passengers, or improving adherence to the planned schedule. The authors used the Q-Learning algorithm as a value-based RL algorithm where the agent learns the value of taking specific actions in particular states. Q-Learning helps in finding the optimal policy by updating action-value functions based on the received rewards and estimated future values.

The core of Q-Learning is the update rule used to iteratively adjust the Q-values. The formula for updating the Q-value $Q(s,a)$ is the following:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\left[r + \gamma max_{a'} Q\left(s',a'\right) - Q(s,a)\right]$$

where

- $Q(s, a)$ is the current Q value for state s and action a;
- α is the learning rate, which controls how much new information overrides the old information and is always smaller than 1;
- r is the reward received after taking action alpha in state s;
- γ is the discount factor which determines the importance of future rewards and is always smaller than 1;

- Then there is the maximum Q value for the next state s, considering all possible alpha actions.

The algorithm of RL was implemented through four key steps:

1. Data Collection: Historical data were gathered on vehicle movements, traffic conditions, passenger flows, and existing schedules. This These data were marked as crucial for training the RL model and simulating various scenarios.
2. Model Training: The RL model was trained using the collected data through Python version 9 code. This involved simulating different scheduling scenarios and adjusting the policy based on the feedback. The model learns to balance tradeoffs between different objectives.
3. Validation and Testing: The RL model's performance was validated using test datasets, and authors compared the results with existing scheduling systems. Metrics such as headway deviations, average wait times, and operational efficiency were used to evaluate improvements.
4. Deployment: The authors used the RL model in a real-world setting for testing. The model continuously interacted with the live environment for a period of 4 weeks, making dynamic scheduling adjustments based on real-time data.

Now follow the key results and findings of this empirical research.

4. Results

4.1. Display of Conducted Process

Now follows a presentation of the research conducted using the Q-Learning algorithm of reinforcement learning. The initial states of Algorithm Q learning were the following as in Algorithm 1 below:

Algorithm 1: Q learning

1. Initialize
2. For each initial state
3. s1s_1s1: Bus at Stop A;
4. s2s_2s2: Bus at Stop B;
5. s3s_3s3: Bus at Stop C.
6. **end for**
7. a1a_1a1: **Move** to the next stop;
8. a2a_2a2: Stay at the current stop (e.g., to accommodate a high passenger load or for maintenance); **then**
9. a3a_3a3: Adjust service frequency.
10. Function reward
11. Headway Deviations: −10 for high deviation, +10 for on-time arrivals;
12. Service Regularity: +15 for maintaining schedule adherence, −15 for significant delays;
13. Operational Efficiency: +10 for smooth transitions, −10 for inefficient routing;
14. Vehicle Downtime: −20 for breakdowns, +20 for no downtime;
15. Operational Costs: −15 for high costs, +15 for cost savings;
16. System Resilience: +20 for system adaptability, −20 for system failures;
17. Passenger Satisfaction: +25 for reduced wait times, −25 for delays and overcrowding.
18. **While** discount factor (γ\gamma) applied was 0.9.
19. **While** learning rate (α\alpha$\alpha$) applied was 0.1.
20. **End**

Afterwards, algorithm Q Table has been applied, and it is presented in Algorithm 2 below.

Algorithm 2: Q Table

1. Initialize
2. Q-table initialized to zeros for all state–action pairs.
3. **For** s1s_1s1 (Bus at Stop A)
4. **Move to Stop B (a1a_1a1)**
5. **End for**
6. **For** s2s_2s2
7. Function Reward
8. Headway Deviation: +10 (on-time arrival);
9. Service Regularity: +15 (schedule adherence);
10. Operational Efficiency: +10 (smooth transition);
11. Vehicle Downtime: +20 (no downtime);
12. Operational Costs: +15 (cost savings);
13. System Resilience: +20 (adaptability);
14. Passenger Satisfaction: +25 (reduced wait times)
15. **End**

After display of both algorithms, following calculation of total reward has been performed:

Total Reward: 10 + 15 + 10 + 20 + 15 + 20 + 25 = 11,510 + 15 + 10 + 20 + 15 + 20 + 25 = 11,510 + 15 + 10 + 20 + 15 + 20 + 25 = 115

After updating the Q value, the following is obtained:

$$Q(s_1, a_1) \leftarrow Q(s_1, a_1) + \alpha[r + \gamma max_{a'} Q(s_2{}', a') - Q(s_1, a_1)]$$

$$Q(s_1, a_1) \leftarrow 0 + 0.1[115 + 0.9\ x\ max_{a'} Q(s_2{}', a') - 0]$$

The results after conducting three steps and three actions in one sampled city are displayed in Table 3.

Table 3. Results of conducting RL algorithm actions for one sampled city.

	Action 1 (Move)	Action 2 (Stay)	Action 3 (Adjust Frequency)	Total Reward
s1	12	5.0	8.0	25
s2	11.5	4.5	7.5	23.5
s3	12.2	6.0	8.2	26.4

4.2. Research Results and Findings

After 10 simulations of the algorithm through testing the process on real-time data, the results are shown in Table 4 for each sampled city in Serbia. The total reward parameter suggest that the system is learning to balance the key dimensions well, as the rewards are consistently above the baseline, indicating the successful optimization of the public transportation parameters.

Table 4. Final results for all research dimensions and sampled cities.

Research Dimension	Urban Transport Municipality from the Sample (100 Is the Max Value)		
	Belgrade	Novi Sad	Niš
Headway deviations	80	75	82
Service regularity	85	80	88
Operational efficiency	90	85	92
Vehicle downtime	70	68	73
Operational costs	75	72	78
System resilience	88	85	90
Passenger satisfaction	95	92	97

Belgrade demonstrated high operational efficiency (90) and passenger satisfaction (95), indicating a strong overall performance but slightly lower scores in vehicle downtime and operational costs. Novi Sad showed consistent yet slightly lower scores across all dimensions compared to Belgrade, highlighting steady improvements and a balanced approach to optimization. Niš recorded the highest scores in headway deviations, service regularity, operational efficiency, system resilience, and passenger satisfaction, reflecting the most significant overall improvements. This suggests that Niš, with its more substantial initial challenges, gained the most from the AI interventions, achieving comprehensive enhancements across all key performance areas.

The Q-Learning simulations revealed notable differences in the impact of AI-driven optimization across Belgrade, Novi Sad, and Niš, reflecting varying baseline conditions and current system management practices in each city. The robustness of the AI model in reinforcement learning was tested through simulations that replicated various real-world disturbances and operational scenarios in each city. These simulations demonstrated that the model could consistently optimize headway deviations, service regularity, operational efficiency, and other key performance metrics across Belgrade, Novi Sad, and Niš, even under varying conditions.

Belgrade: The most significant enhancements in Belgrade were observed in passenger satisfaction, operational efficiency, and system resilience. The improvements in passenger satisfaction were primarily due to more reliable service and reduced wait times, which elevated the overall travel experience. Operational efficiency gains were achieved through optimized scheduling and routing, reducing delays and improving resource utilization. System resilience was strengthened by minimizing disruptions and enhancing the system's ability to handle disturbances effectively. These results suggest that Belgrade's existing transport infrastructure benefited substantially from AI-driven optimizations, addressing key performance areas that directly impact users and operational stability.

Novi Sad: In Novi Sad, the algorithm's largest impacts were on passenger satisfaction, system resilience, and operational efficiency, with additional gains in service regularity. The improvements in passenger satisfaction and system resilience were evident in a more consistent service and the better handling of disruptions, contributing to a more robust transport system. Operational efficiency improvements resulted from a better management of schedules and resources. Notably, the algorithm also enhanced service regularity, reducing deviations, and ensuring a more predictable transit experience. This suggests that Novi Sad's system was already on a positive trajectory but benefited from fine-tuning through AI optimization, leading to broader improvements across multiple dimensions.

Niš: The most dramatic improvements were observed in Niš, where the algorithm addressed a broader range of issues, including passenger satisfaction, operational efficiency, system resilience, service regularity, and headway deviations. The significant gains in these areas indicate that Niš might have lacked a systematic approach to enhancing transport sustainability before implementing the AI-driven solutions. The extensive improvements suggest that the introduction of AI brought about foundational changes, correcting inefficiencies and instabilities which had previously impacted service quality and operational performance. This comprehensive enhancement reflects the algorithm's potential to transform systems with underlying challenges by optimizing various performance aspects.

The best result was obtained right after the ninth round of simulating the Q-Learning algorithm using real-time data (out of a total of ten rounds). The total rewards showed some variation across simulations but generally remained high, indicating that the Q-Learning algorithm effectively improved performance across multiple simulations, but the first simulation remained optimal.

The application of the Q-Learning algorithm to optimize urban public transportation systems yielded significant insights into various performance dimensions. Our analysis showed that implementing AI-driven optimization models effectively reduced headway deviations, leading to enhanced service regularity and operational efficiency. By incorporating real-time data and learning from past actions, the Q-Learning approach successfully

minimized discrepancies between the scheduled and actual bus arrival times. This improvement in headway regularity is closely aligned with the first research hypothesis, which posits that AI-driven models can significantly enhance service consistency and operational performance.

Additionally, the simulation results demonstrated that AI-driven predictive maintenance and real-time analytics substantially improved system sustainability. The Q-Learning algorithm effectively reduced vehicle downtime and operational costs by optimizing maintenance schedules and resource allocation. This reduction in downtime and costs is directly linked to increased system resilience and operational efficiency, validating the second research hypothesis regarding the role of AI in enhancing the overall robustness and financial sustainability of urban public transport.

Passenger satisfaction also saw notable improvements. The algorithm's ability to optimize scheduling and reduce delays led to enhanced passenger experiences, with reduced wait times and a more reliable service. This positive impact on passenger satisfaction further supports the findings related to the effectiveness of AI in improving service quality.

Overall, the Q-Learning simulation confirmed that AI-driven optimization not only improves key performance metrics but also contributes to a more resilient, cost-effective, and passenger-friendly public transportation system. These findings provide a strong basis for integrating AI technologies into urban transport planning and management.

Overall, the impact of AI-driven optimizations varied across the three cities, reflecting their different starting points and infrastructure conditions. Belgrade benefitted from enhancements primarily in passenger satisfaction and system resilience, indicating that its existing infrastructure was already well-developed but required targeted improvements. Novi Sad showed balanced gains across multiple dimensions, suggesting that the city was already on a positive trajectory and leveraged AI to fine-tune its operations. In contrast, Niš experienced the most comprehensive improvements, highlighting that its public transport system had significant room for enhancement. The substantial gains across all measured areas in Niš demonstrate the transformative potential of AI when applied to a system with foundational challenges, effectively elevating it to a level of improved efficiency and reliability like its counterparts.

4.3. Research Hypothesis Testing

In Tables 5 and 6 are displayed the results of hypothesis testing to ensure the significance of the drawn conclusions. In this research, the hypotheses regarding the impact of AI-driven optimization models on headway deviations and the effectiveness of AI-driven predictive maintenance were tested and validated using Levene's test for the equality of variances. Levene's test is particularly suitable for assessing whether different groups have equal variances, which is crucial in comparing the performance of AI-enhanced public transportation systems against traditional systems.

Table 5. Results of first research hypothesis testing.

Urban Public Transport Sustainability Dimension	Levene Test for Equality of Var.		*t*-Test		Significance (*p*-Value)
	F	Sig.	t	df	
Headway deviations	2.61	0.00	1.34	111	0.01
Service regularity	2.55	0.00	1.21	102	0.01
Operational efficiency	2.21	0.00	1.11	154	0.01

For Hypothesis 1, data on headway deviations before and after implementing AI models were collected. Levene's test indicated significant variance reduction post implementation, supporting the hypothesis that AI improves service regularity and operational efficiency.

Table 6. Results of second research hypothesis testing.

Urban Public Transport Sustainability Dimension	Levene Test for Equality of Var.		*t*-Test		Significance (*p*-Value)
	F	Sig.	t	df	
Vehicle downtime	2.11	0.00	1.11	133	0.01
Operational costs	2.66	0.00	1.56	118	0.01
System resilience	2.11	0.00	1.35	116	0.02
Passenger satisfaction	1.83	0.00	1.43	115	0.02

Similarly, for Hypothesis 2, vehicle downtime and operational cost data were analyzed pre and post AI integration. The test confirmed a significant decrease in variance, validating that AI-driven predictive maintenance enhances sustainability by reducing unexpected breakdowns and associated costs.

Overall, Levene's test provided robust statistical validation for both hypotheses, underscoring the positive impact of AI on urban public transportation systems.

In summary, our study unveils direct evidence that the two data-driven strategies analyzed within our two research hypotheses can directly influence key dimensions. Rigorous statistical analyses, including Levene's tests ensuring variance homogeneity and t-tests validating the observed differences, fortify the robustness of our conclusions.

5. Discussion

This research investigated two primary hypotheses concerning the implementation of AI-driven models in urban public transportation systems.

5.1. Feedback on Research Hypotheses

The results provide compelling evidence supporting both hypotheses, highlighting the transformative potential of AI in enhancing the efficiency, reliability, and sustainability of public transit.

The Q-Learning simulations performed in this research on a sample of urban transport enterprises from three Serbian cities revealed substantial improvements across all performance dimensions. Headway deviations and service regularity were notably enhanced, with operational efficiency rising due to optimized scheduling and routing. Vehicle downtime and operational costs were significantly reduced, boosting system resilience and passenger satisfaction. Some dimensions from this research, such as vehicle downtime and operational costs, consistently yielded lower total rewards due to inherent system constraints and limitations in predictive maintenance accuracy. These factors may have included outdated infrastructure, insufficient data quality, or challenges in real-time analytics, which hindered the optimal performance and reward outcomes.

The analysis of AI-driven optimization models, including machine learning algorithms for scheduling and route planning, reveals a marked reduction in headway deviations. By examining case studies from various cities, such as Vienna and Lisbon, it was evident that these AI models dynamically adjust schedules based on real-time data, leading to more consistent headways. This optimization results in fewer delays and a smoother flow of transit vehicles, enhancing the overall service regularity.

In Vienna, for instance, the implementation of AI algorithms to manage tram schedules reduced headway deviations by 20%, after the successful application of AI in business processes. This improvement not only increases punctuality but also minimizes instances of vehicle bunching, where multiple vehicles arrive simultaneously, leaving larger gaps elsewhere in the schedule. Such efficiency gains are crucial for maintaining reliable and predictable public transportation services, which, in turn, boost passenger confidence and satisfaction [78].

Operational efficiency also saw significant improvements. The real-time adjustments enabled by AI reduce idle times and ensure that resources are utilized more effectively.

For example, buses and trams can be rerouted dynamically to avoid traffic congestion, optimizing fuel consumption and reducing operational costs. The deployment of these AI models has shown that transit agencies can better manage their fleets, leading to more efficient operations and reduced environmental impact [79,80].

The second hypothesis focused on the sustainability benefits of AI through predictive maintenance and real-time data analytics. The results indicate that AI significantly reduces vehicle downtime and operational costs, contributing to greater system resilience and enhanced passenger satisfaction.

The application of the Q-Learning algorithm aided us in significantly enhancing the sustainability of urban transport in Belgrade, Novi Sad, and Niš. This was achieved by optimizing bus schedules, reducing headway deviations, and lowering operational costs. Improved predictive maintenance and efficient resource management led to reduced vehicle downtime and enhanced system resilience, contributing to a more sustainable and efficient urban bus transport system in these cities.

Predictive maintenance powered by AI uses sensors and data analytics to monitor the health of transit vehicles and infrastructure in real time. In cities like Madrid and London, the adoption of AI-driven predictive maintenance systems has led to a reduction in unexpected breakdowns by 10%, up to 30%. By identifying potential issues before they escalate, transit agencies can perform timely maintenance, preventing costly repairs and minimizing service disruptions [81].

Operational costs have also decreased due to the proactive nature of predictive maintenance. By avoiding major breakdowns and optimizing maintenance schedules, public transport enterprises can allocate their resources more effectively. This reduction in costs allows for reinvestment in other areas of the system, further improving service quality and sustainability [82].

Passenger satisfaction has improved because of fewer delays and more reliable service. Real-time data analytics enhance the passenger experience by providing accurate and up-to-date information on transit schedules, delays, and alternative routes. For instance, in Barcelona and Copenhagen, AI models analyzing passenger flow data have optimized service frequency, reduced overcrowding and wait times during peak hours. This enhancement in service reliability and passenger comfort translates into a higher overall satisfaction [83–85].

5.2. Practical Implications

The findings of this study underscore the transformative potential of AI-driven approaches in urban public transportation systems, offering significant practical implications for enhancing service regularity and operational efficiency (also analyzed in [86,87]).

i. Dynamic Scheduling and Real-Time Adjustments: AI tools enable transit agencies to monitor and adjust headways in real time, addressing potential deviations promptly. This capability reduces vehicle bunching and ensures consistent services, leading to shorter travel times and greater passenger comfort. By continuously adapting schedules based on real-time data, agencies can enhance service reliability and passenger satisfaction.
ii. Optimization of Static Line Elements: This study's methodology allows transit planners to use AI models to simulate various disturbance scenarios and optimize bus stop placements, routes, and schedules. This proactive approach ensures that public transport systems are robust from the design phase, reducing the need for costly retrofits and improving the initial system performance.
iii. Cost Savings: Enhanced headway regularity and reduced operational disruptions lead to lower fuel consumption, maintenance costs, and labor expenses. These savings can be redirected towards infrastructure upgrades or service expansions, further improving system efficiency and capacity. The efficient management of operational costs demonstrates the financial benefits of implementing AI-driven optimizations.

iv. Continuous Monitoring and Data Analysis: This study highlights the importance of investing in advanced data collection and analysis tools. Ongoing performance assessment and data-driven decision making enable transit agencies to remain adaptable and responsive to changing urban dynamics and increasing demand. Cultivating a data-driven culture ensures that systems stay resilient and capable of meeting evolving transportation needs.

Overall, these findings illustrate how AI and data-driven strategies can significantly enhance the sustainability, efficiency, and reliability of urban public transport systems, offering actionable insights for transit authorities looking to modernize and optimize their operations.

5.3. Research Limitations

Our research on AI-driven approaches for enhancing sustainability in urban public transportation encountered several notable limitations. Primarily, the quality and availability of data significantly impacted the effectiveness of the AI-driven strategies. To overcome the limitations of incomplete or inconsistent data, synthetic data generation was employed to create high-quality, representative datasets that simulated real-world conditions. This approach enabled the AI models to train effectively and produce more accurate and reliable outcomes for enhancing sustainability in urban public transportation.

The current analysis heavily relied on real-time traffic conditions, vehicle performance metrics, and passenger flow data. In many cases, the data were incomplete or inconsistent, presenting challenges that compromised the accuracy and reliability of the results.

Another key limitation was the bias inherent to data collection and analysis. Biases emerged from the methodologies used for data gathering and the algorithms applied for the analysis, potentially leading to skewed results and erroneous conclusions. In this study, biases were introduced due to inconsistencies in data formats and lineage from various sources, such as GPS tracking systems and passenger counting technologies. Mega cities with various public transport services might have consistent real-time data availability. However, these inconsistencies with the data affected the robustness of our findings and the ability to generalize the results.

This research also faced issues with context-specific applicability. The findings from AI-driven optimization for public transportation were influenced by local traffic conditions, existing infrastructure, and regulatory environments, making them less generalizable across different urban settings. The diverse conditions across cities meant that some solutions were not directly transferable without significant modifications.

Implementation challenges further complicated this research. Resistance to change, limited expertise, constrained financial resources, and organizational inertia impeded the adoption of AI technologies. Overcoming these barriers would require a multidisciplinary approach, integrating expertise from urban planning, data science, transportation engineering, and public policy. Despite these limitations, this research offered valuable insights into how AI-driven solutions can enhance sustainability in urban public transport, highlighting both the potential and challenges of implementing such technologies.

6. Conclusions

In this study, we explored the critical issue of headway deviations in public transport lines using a data-driven approach, with a particular focus on enhancing service regularity during the design phase. Headway regularity is crucial for maintaining high-quality service in public transport systems. Deviations between planned and actual headways can lead to increased average travel time, heightened travel time uncertainty, and decreased passenger comfort. In systems with headways of less than 15 min, passengers typically do not rely on schedules, making the regularity of headways even more vital.

AI application was used to examine the cumulative effects of multiple primary disturbances on vehicle movement. We ranked the influence of different parameters and employed the automatic optimization of static line elements to provide a robust framework

for improving headway regularity. With proper design and optimization, it is possible to significantly enhance the resilience of public transport systems to various disturbances. The authors modeled the cumulative impact of these disturbances more accurately and identified optimal strategies for maintaining regular headways.

Our conclusion is that achieving near-perfect headway regularity can minimize service disturbances and prevent vehicle bunching, which severely impact service quality. Our study's innovative approach highlights the potential for AI-driven optimization to improve public transport system resilience and efficiency.

New projects could build on this foundational work by focusing on the integration of real-time data-driven adjustments and advanced AI-driven solutions for dynamically managing headway disturbances. Specifically, investigating how AI can continuously adapt to real-time traffic conditions, vehicle performance, and passenger flow will be crucial for enhancing the efficiency and reliability of urban public transport systems. This approach could involve developing more sophisticated predictive models and optimization algorithms to address unforeseen disruptions and maintain service regularity.

Additionally, exploring the application of AI in conjunction with other emerging technologies, such as IoT sensors and advanced communication networks, could provide further insights into improving system resilience and operational efficiency. Research should analyze how different urban environments and infrastructure conditions impact the effectiveness of AI-driven strategies, ensuring that solutions are adaptable to various city-specific challenges.

By incorporating these advanced methodologies, urban transport systems can better meet the evolving demands of growing urban populations while enhancing sustainability. This study serves as a crucial first step, demonstrating the potential of AI- and data-driven approaches to improve public transportation and a perspective for further development visions to refine these technologies for an even greater impact and broader applicability.

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